

Statement of Work

Project: Acquisition of cell analyzer instrument for enumeration and characterization of animal cells and nuclei

GENERAL INFORMATION

1.0 Scope of Work:

Acquisition of cell analyzer instrument for enumeration of non-viable and viable cells and nuclei and simultaneous acquisition of light scatter properties (forward and side scatter) and detection of fluorescent labels.

2.0 Background:

The research unit requires a system that can enumerate eukaryotic cells (primary cells and cell lines) or nuclei isolated from eukaryotic cells in a suspension created from various animal tissues for downstream assays. The system must determine the number of non-viable and viable cells in a sample in minimal time and with limited input and interpretation from the operator (high level of automation). Acquisition of light scatter characteristics and detection of fluorescent labels in the side scatter channel (SSC) are required to assess cell and nuclei characteristics, which is required for subsequent assays. The instrument needs to be small, portable, rapid, and require minimal sample and operator input. Additionally, an instrument that can rapidly identify populations via 5 different channels is required (forward scatter channel, side scatter channel, and 3 fluorescent based channels) for identification of cell/nuclei integrity, stress parameters, and user defined cell populations.

Some cell and nuclei enumeration instruments or methods require operator identification of live and total events with laborious calculations required to determine percent viability and concentration. Some instruments are unable to accurately identify small particles or nuclei and often mislabel small cells as debris or dead cells, especially in some food animal species (avians). Large footprint instruments are unable to be readily moved into and out of the biosafety cabinet readily when working with biological hazardous samples. Thus, an instrument that is small, portable, capable of accurately counting a wide range of live eukaryotic cells or intact nuclei with minimal sample and operator input is required.

CONTRACTOR REQUIREMENTS

3.0 Technical Requirements/Tasks:

- 1) Instrument must use laser-based fluorescence detection via microcapillary system for detection and enumeration of cells or nuclei and analysis with a 5-decade dynamic range.
- 2) With maximum accuracy (average % CV less than 5% for concentration and viability) the instrument must determine cell counts, cell viability, cell death (apoptosis) and cell cycle with 5 parameter fluorescent-based detection.
- 3) Instrument must allow for detection of multiple fluorescent molecules with minimal spectral overlap (minimum 3) as well as forward scatter and side scatter properties.
- 4) Instrument must have rapid processing time once sample is loaded into instrument (< 1 minute per sample is preferred; up to 2 min maximum).

- 5) Instrument must report total cells in sample, percent live cells, total live cells, live cells/ml, and total cells/ml with minimal input from operator.
- 6) Instrument must be small, portable, with minimal set up to allow use in a biosafety cabinet (< 15lb, fit in biological safety cabinet (max of 10in x 15in x 10 in – width x depth x height).
- 7) Instrument must include a defined set up that includes quality control steps for accurate counting and manufacturer recommended protocols for effective cleaning before and/or after each use.
- 8) Computer system must be built into the instrument to minimize footprint on bench and enhance ability to decontaminate external components of instrument. Internal components must allow for decontamination of fluidics using bleach or comparable disinfectant solution.
- 9) Fully optimized software modules must be built into the instrument for any reagents provided by the manufacturer (purchased separately) to allow full analysis of data in real time with accuracy and intuitive analysis. Data files must be output in Flow Cytometry Standard (FCS) 3.0 and comma-separated value (CSV) format at a minimum, but instrument should also display results in real time either in summary format or in representative dot plots. Exporting data for record retention through a flash drive is sufficient; cloud-based output methods are not desired.
- 10) Instrument must include built in software that allows for user to define assays for 1-3 fluorescent labels with minimal spectral overlap. Fluorescent channels detected must include FITC, PE, and PE-Cy5 or more specifically, green fluorescence centered at 525nm, yellow fluorescence at 576nm, and red fluorescence at 680nm (all detected within 30nm of the specificized centered bandwidth). This is in addition to detection of forward and side scatter properties.
- 11) Instrument must operate with a minimum sample volume of 200ul and a sample dead volume of 50ul.
- 12) Instrument shall have capabilities of detecting eukaryotic cells of multiple sizes or nuclei of different sizes (approximately 0.5-60 um in diameter). Ability to enumerate bacteria is not required but will be considered if available.
- 13) Vendor must offer one-step reagents (purchased separately) to label cells for enumeration of viable cells and reagents for cell counting and cell viability assessment that will provide absolute cell count and viability data on various cell suspensions. Reagents must differentially stain cells based on permeability to the DNA-binding dyes in the reagent.
- 14) Instrument must be a new, unused product with minimum one year warranty.
- 15) Prefer the following items. Name brand or equal. Qty: 1 of each.

Cytek Muse Micro Blue 5 Channel N7-00142
 Muse System Check Kit MCH100101
 Guava Instr Clean Fluid 100 ml 4200-0140
 Include shipping costs to the location.

4.0 Government Furnished

No access to the building is required; user should be able to assemble and set up upon delivery.

5.0 Deliverables/Schedule:

Complete system to be delivered to:

USDA-NADC
1920 Dayton Avenue
Bldg 24
Ames, IA 50010

6.0 Evaluation Factors

Technical acceptability is more important than price.

7.0 Technical Point of Contact

Should there be questions, contact James Porter at james.porter@usda.gov.